

Loss of microglial SIRP α promotes synaptic pruning in preclinical models of neurodegeneration.

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Microglia play a key role in regulating synaptic remodeling in the central nervous system. Activation of classical complement pathway promotes microglia-mediated synaptic pruning during development and disease. CD47 protects synapses from excessive pruning during development, implicating microglial SIRP α , a CD47 receptor, in synaptic remodeling. However, the role of microglial SIRP α in synaptic pruning in disease remains unclear. Here, using conditional knock-out mice, we show that microglia-specific deletion of SIRP α results in decreased synaptic density. In human tissue, we observe that microglial SIRP α expression declines alongside the progression of Alzheimer's disease. To investigate the role of SIRP α in neurodegeneration, we modulate the expression of microglial SIRP α in mouse models of Alzheimer's disease. Loss of microglial SIRP α results in increased synaptic loss mediated by microglia engulfment and enhanced cognitive impairment. Together, these results suggest that microglial SIRP α regulates synaptic pruning in neurodegeneration.

Loss of microglial MCT4 leads to defective synaptic pruning and anxiety-like behavior in mice.

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Microglia, the innate immune cells of the central nervous system, actively participate in brain development by supporting neuronal maturation and refining synaptic connections. These cells are emerging as highly metabolically flexible, able to oxidize different energetic substrates to meet their energy demand. Lactate is particularly abundant in the brain, but whether microglia use it as a metabolic fuel has been poorly explored. Here we show that microglia can import lactate, and this is coupled with increased lysosomal acidification. In vitro, loss of the monocarboxylate transporter MCT4 in microglia prevents lactate-induced lysosomal modulation and leads to defective cargo degradation. Microglial depletion of MCT4 in vivo leads to impaired synaptic pruning, associated with increased excitation in hippocampal neurons, enhanced AMPA/GABA ratio, vulnerability to seizures and anxiety-like phenotype. Overall, these findings show that selective disruption of the MCT4 transporter in microglia is sufficient to alter synapse refinement and to induce defects in mouse brain development and adult behavior.

Dark Microglia Are Abundant in Normal Postnatal Development, where they Remodel Synapses via Phagocytosis and Trophocytosis, and Are Dependent on TREM2.

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This study examined dark microglia—a state linked to central nervous system pathology and neurodegeneration—during postnatal development in the mouse ventral hippocampus, finding that dark microglia interact with blood vessels and synapses and perform trophocytosis of pre-synaptic axon terminals. Furthermore, we found that dark microglia in development notably expressed C-type lectin domain family 7 member A (CLEC7a), lipoprotein lipase (LPL) and triggering receptor expressed on myeloid cells 2 (TREM2) and required TREM2, differently from other microglia, suggesting a link between their role in remodeling during development and central nervous system pathology. Together, these

results point towards a previously under-appreciated role for dark microglia in synaptic pruning and plasticity during normal postnatal development.

Glial Cell Modulation of Dendritic Spine Structure and Synaptic Function.

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Glia comprise a heterogeneous group of cells involved in the structure and function of the central and peripheral nervous system. Glial cells are found from invertebrates to humans with morphological specializations related to the neural circuits in which they are embedded. Glial cells modulate neuronal functions, brain wiring and myelination, and information processing. For example, astrocytes send processes to the synaptic cleft, actively participate in the metabolism of neurotransmitters, and release gliotransmitters, whose multiple effects depend on the targeting cells. Human astrocytes are larger and more complex than their mice and rats counterparts. Astrocytes and microglia participate in the development and plasticity of neural circuits by modulating dendritic spines. Spines enhance neuronal connectivity, integrate most postsynaptic excitatory potentials, and balance the strength of each input. Not all central synapses are engulfed by astrocytic processes. When that relationship occurs, a different pattern for thin and large spines reflects an activity-dependent remodeling of motile astrocytic processes around presynaptic and postsynaptic elements. Microglia are equally relevant for synaptic processing, and both glial cells modulate the switch of neuroendocrine secretion and behavioral display needed for reproduction. In this chapter, we provide an overview of the structure, function, and plasticity of glial cells and relate them to synaptic maturation and modulation, also involving neurotrophic factors. Together, neurons and glia coordinate synaptic transmission in both normal and abnormal conditions. Neglected over decades, this exciting research field can unravel the complexity of species-specific neural cytoarchitecture as well as the dynamic region-specific functional interactions between diverse neurons and glial subtypes.

Lifelong absence of microglia alters hippocampal glutamatergic networks but not synapse and spine density.

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Microglia sculpt developing neural circuits by eliminating excess synapses in a process called synaptic pruning, by removing apoptotic neurons, and by promoting neuronal survival. To elucidate the role of microglia during embryonic and postnatal brain development, we used a mouse model deficient in microglia throughout life by deletion of the *fms*-intronic regulatory element (FIRE) in the *Csf1r* locus. Surprisingly, young adult *Csf1r* Δ FIRE/ Δ FIRE mice display no changes in excitatory and inhibitory synapse number and spine density of CA1 hippocampal neurons compared with *Csf1r* $^{+/+}$ littermates. However, CA1 neurons are less excitable, receive less CA3 excitatory input and show altered synaptic properties, but this does not affect novel object recognition. Cytokine profiling indicates an anti-inflammatory state along with increases in ApoE levels and reactive astrocytes containing synaptic markers in *Csf1r* Δ FIRE/ Δ FIRE mice. Notably, these changes in *Csf1r* Δ FIRE/ Δ FIRE mice closely resemble the effects of acute microglial depletion in adult mice after normal development. Our findings suggest that microglia are not mandatory for synaptic pruning, and that in their absence pruning can be achieved by other mechanisms.

Impaired microglia-mediated synaptic pruning in the nucleus accumbens during adolescence results in persistent dysregulation of familiar, but not novel social interactions in sex-specific ways.

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Evolutionarily conserved, peer-directed social behaviors are essential to participate in many aspects of human society. These behaviors directly impact psychological, physiological, and behavioral maturation. Adolescence is an evolutionarily conserved period during which reward-related behaviors, including social behaviors, develop via developmental plasticity in the mesolimbic dopaminergic 'reward' circuitry of the brain. The nucleus accumbens (NAc) is an intermediate reward relay center that develops during adolescence and mediates both social behaviors and dopaminergic signaling. In several developing brain regions, synaptic pruning mediated by microglia, the resident immune cells of the brain, is important for normal behavioral development. In rats, we previously demonstrated that microglial synaptic pruning also mediates NAc and social development during sex-specific adolescent periods and via sex-specific synaptic pruning targets. In this report, we demonstrate that interrupting microglial pruning in NAc during adolescence persistently dysregulates social behavior towards a familiar, but not novel social partner in both sexes, via sex-specific behavioral expression. This leads us to infer that naturally occurring NAc pruning serves to reduce social behaviors primarily directed toward a familiar conspecific in both sexes, but in sex-specific ways.

Commentary: How Do Microglia Regulate Neural Circuit Connectivity and Activity in the Adult Brain?

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Microglia are the primary immune cells in CNS. Recent work shows that microglia are also essential for proper brain development through synaptic pruning and remodeling during early life development. But the question of whether and how microglia regulate synaptic connectivity in the adult brain remains open. Our recently published study provides new insights into the functional roles of microglia in the adult mouse brain. We find that chronic depletion of microglia via CSF1R inhibitors in the visual cortex in adult mice induces a dramatic increase in perineuronal nets, and enhances neural activities of both excitatory neurons and parvalbumin interneurons. These findings highlight new potential therapeutic avenues to enhance adult neural plasticity by manipulating microglia.

N-acetylaspartate mitigates pro-inflammatory responses in microglial cells by intersecting lipid metabolism and acetylation processes.

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Microglia play a crucial role in brain development and repair by facilitating processes such as synaptic pruning and debris clearance. They can be activated in response to various stimuli, leading to either pro-inflammatory or anti-inflammatory responses associated with specific metabolic alterations. The imbalances between microglia activation states contribute to chronic neuroinflammation, a hallmark of

neurodegenerative diseases. N-acetylaspartate (NAA) is a brain metabolite predominantly produced by neurons and is crucial for central nervous system health. Alterations in NAA metabolism are observed in disorders such as Multiple Sclerosis and Canavan disease. While NAA's role in oligodendrocytes and astrocytes has been investigated, its impact on microglial function remains less understood. The murine BV2 microglial cell line and primary microglia were used as experimental models. Cells were treated with exogenous NAA and stimulated with LPS/IFN- γ to reproduce the pro-inflammatory phenomenon. HPLC and immunofluorescence analysis were used to study lipid metabolism following NAA treatment. Automated fluorescence microscopy was used to analyze phagocytic activity. The effects on the pro-inflammatory response were evaluated by analysis of protein/mRNA expression and ChIP assay of typical inflammatory markers. NAA treatment promotes an increase in both lipid synthesis and degradation, and enhances the phagocytic activity of BV2 cells, thus fostering surveillant microglia characteristics. Importantly, NAA decreases the pro-inflammatory state induced by LPS/IFN- γ via the activation of histone deacetylases (HDACs). These findings were validated in primary microglial cells, highlighting the impact on cellular metabolism and inflammatory responses. The study highlighted the role of NAA in reinforcing the oxidative metabolism of surveillant microglial cells and, most importantly, in buffering the inflammatory processes characterizing reactive microglia. These results suggest that the decreased levels of NAA observed in neurodegenerative disorders can contribute to chronic neuroinflammation.

Microglia and Astrocytes in Postnatal Neural Circuit Formation.

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Over the past two decades, microglia and astrocytes have emerged as critical mediators of neural circuit formation. Particularly during the postnatal period, both glial subtypes play essential roles in orchestrating nervous system development through communication with neurons. These functions include regulating synapse elimination, modulating neuronal density and activity, mediating synaptogenesis, facilitating axon guidance and organization, and actively promoting neuronal survival. Despite the vital roles of both microglia and astrocytes in ensuring homeostatic brain development, the extent to which the postnatal functions of these cells are regulated by sex and the manner in which these glial cells communicate with one another to coordinate nervous system development remain less well understood. Here, we review the critical functions of both microglia and astrocytes independently and synergistically in mediating neural circuit formation, focusing our exploration on the postnatal period from birth to early adulthood.

Loss of C1q alters the auditory brainstem response.

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Neural circuits in the auditory brainstem compute interaural time and intensity differences used to determine the locations of sound sources. These circuits display features that are specialized for these functions. The projection from the ventral cochlear nucleus (VCN) to the medial nucleus of the trapezoid (MNTB) body travels along highly myelinated fibers and terminates in the calyx of Held. This monoinnervating synapse emerges during development as multiple inputs are eliminated. We previously demonstrated that elimination of microglia with a colony stimulating factor-1 inhibitor results in impaired synaptic pruning so that multiple calyceal terminals reside on principal cells of MNTB. This inhibitor also

resulted in impaired auditory brainstem responses (ABRs), with elevated thresholds and increased peak latencies. Loss of the microglial fractalkine receptor, CX3CR1, decreased peak latencies in the ABR. The mechanisms underlying these effects are not known. One prominent microglial signaling pathway involved in synaptic pruning and plasticity during development and aging is the C1q-initiated complement cascade. Here we investigated the classical complement pathway initiator, C1q, in auditory brainstem maturation. We found that C1q expression is detected in the MNTB by the first postnatal week. C1q levels increased with age and were detected within microglia and surrounding the soma of MNTB principal neurons. Loss of C1q did not affect microglia-dependent calyceal pruning. Excitatory and inhibitory synaptic markers in the MNTB and LSO were not altered with C1q deletion. ABRs showed that C1q KO mice had normal hearing thresholds but shortened peak latencies. Altogether this study uncovers the developmental time frame of C1q expression in the sound localization pathway and shows a subtle functional consequence of C1q knockdown.
